

Theorem (Jensen's inequality). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex. For any integer $n \geq 1$, any real numbers $x_1, \dots, x_n$ and any weights $\lambda_1, \dots, \lambda_n$ satisfying*

$$\lambda_i \geq 0 \ (i = 1, \dots, n), \quad \sum_{i=1}^n \lambda_i = 1,$$

we have

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i). \quad (1)$$

Equality holds exactly when either all points with positive weight are identical, or f is affine on the convex hull of those points.

Proof. **Binary convexity.** Convexity of f is equivalent to

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) \quad (x, y \in \mathbb{R}, t \in [0, 1]). \quad (C)$$

Base case $n = 1$. If $n = 1$, then the only admissible weight is $\lambda_1 = 1$, so

$$f\left(\sum_{i=1}^1 \lambda_i x_i\right) = f(x_1) = \sum_{i=1}^1 \lambda_i f(x_i),$$

and (1) holds with equality.

Induction hypothesis. Assume that for some $k \geq 1$ inequality (1) holds for every choice of $x_1, \dots, x_k$ and non-negative weights $\lambda_1, \dots, \lambda_k$ with $\sum_{i=1}^k \lambda_i = 1$:

$$f\left(\sum_{i=1}^k \lambda_i x_i\right) \leq \sum_{i=1}^k \lambda_i f(x_i). \quad (\text{IH})$$

Inductive step $k \rightarrow k + 1$. Let $\lambda_1, \dots, \lambda_{k+1} \geq 0$ with $\sum_{i=1}^{k+1} \lambda_i = 1$ and let $x_1, \dots, x_{k+1} \in \mathbb{R}$.

Set

$$\mu := \sum_{i=1}^k \lambda_i, \quad \lambda_{k+1} = 1 - \mu.$$

If $\mu = 0$ then $\lambda_{k+1} = 1$ and we are back to the base case, so (1) is true. Henceforth assume $\mu > 0$.

Define normalized weights for the first k points,

$$\alpha_i := \frac{\lambda_i}{\mu} \quad (i = 1, \dots, k),$$

and the intermediate convex combination

$$y := \sum_{i=1}^k \alpha_i x_i = \frac{1}{\mu} \sum_{i=1}^k \lambda_i x_i.$$

Clearly $\alpha_i \geq 0$ and $\sum_{i=1}^k \alpha_i = 1$.

Step 1 – binary convexity. Apply (??) with $x := y$, $y := x_{k+1}$ and $t := \mu \in [0, 1]$:

$$f(\mu y + (1 - \mu)x_{k+1}) \leq \mu f(y) + (1 - \mu)f(x_{k+1}). \quad (2)$$

Since $\mu y + (1 - \mu)x_{k+1} = \sum_{i=1}^k \lambda_i x_i + \lambda_{k+1} x_{k+1} = \sum_{i=1}^{k+1} \lambda_i x_i$, inequality (??) becomes

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) \leq \mu f(y) + (1 - \mu)f(x_{k+1}). \quad (3)$$

Step 2 – induction hypothesis on the first k points. Applying (IH) to $x_1, \dots, x_k$ with weights $\alpha_1, \dots, \alpha_k$ yields

$$f(y) = f\left(\sum_{i=1}^k \alpha_i x_i\right) \leq \sum_{i=1}^k \alpha_i f(x_i). \quad (4)$$

Multiplying (4) by μ and using $\mu\alpha_i = \lambda_i$ gives

$$\mu f(y) \leq \sum_{i=1}^k \lambda_i f(x_i). \quad (5)$$

Step 3 – conclusion of the inductive step. Insert (5) into (3):

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) \leq \sum_{i=1}^k \lambda_i f(x_i) + (1 - \mu)f(x_{k+1}) = \sum_{i=1}^{k+1} \lambda_i f(x_i),$$

which is precisely (1) for $n = k + 1$.

Thus the inequality holds for all $n \geq 1$ by induction.

Equality cases. In the binary convexity (??) equality occurs iff either $t = 0$ or $t = 1$ (one of the two weights is zero) or f is affine on the segment joining the two points. Equality in the induction hypothesis (IH) occurs iff either all points with positive weight among $x_1, \dots, x_k$ coincide or f is affine on their convex hull.

Combining these observations through the inductive construction shows that equality in (??) holds exactly when

[(i)]

1. all points x_i with $\lambda_i > 0$ are identical, or
2. f is affine on the convex hull $\text{conv}\{x_i : \lambda_i > 0\}$.

These are the classical equality conditions for Jensen's inequality. □ □